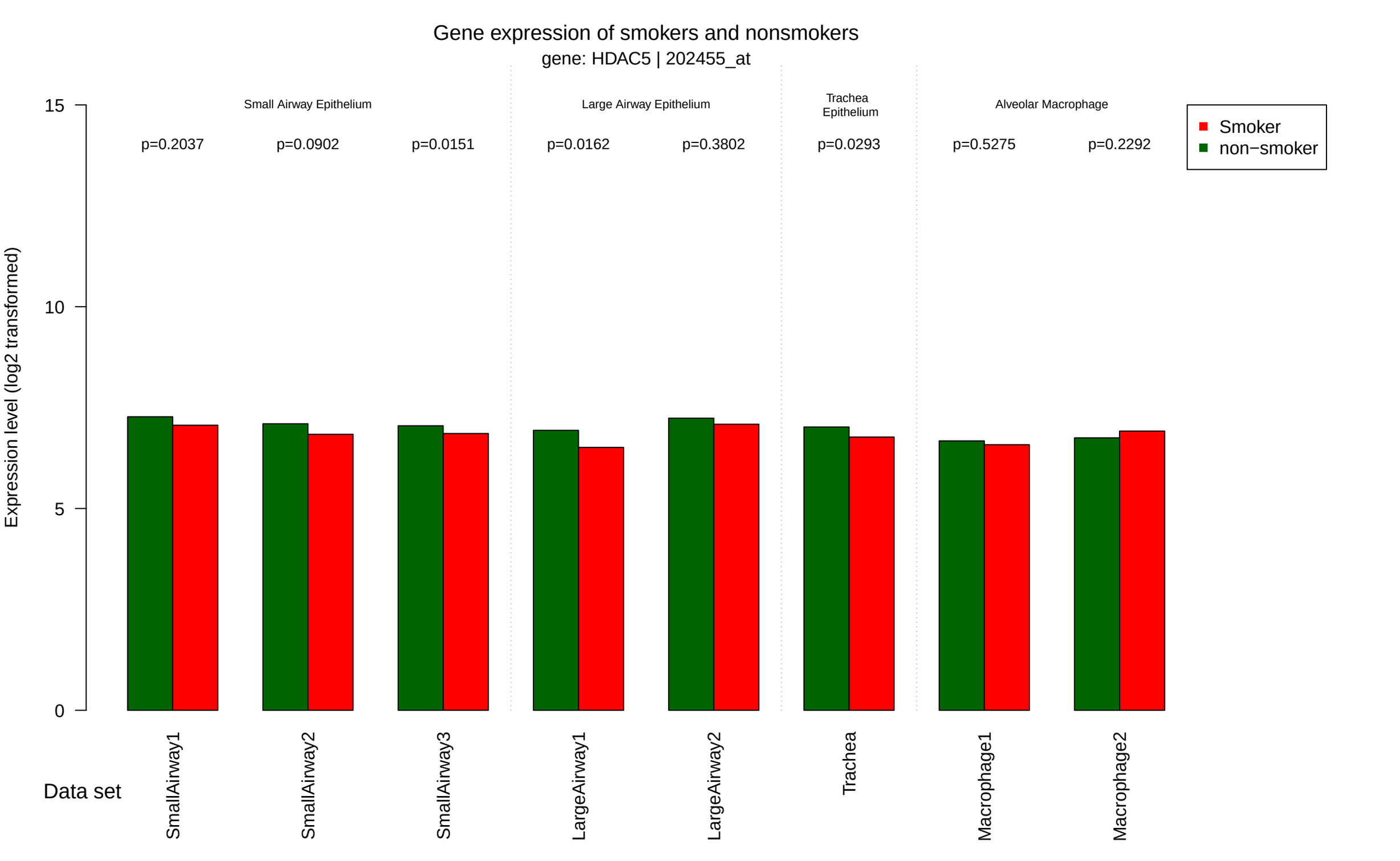
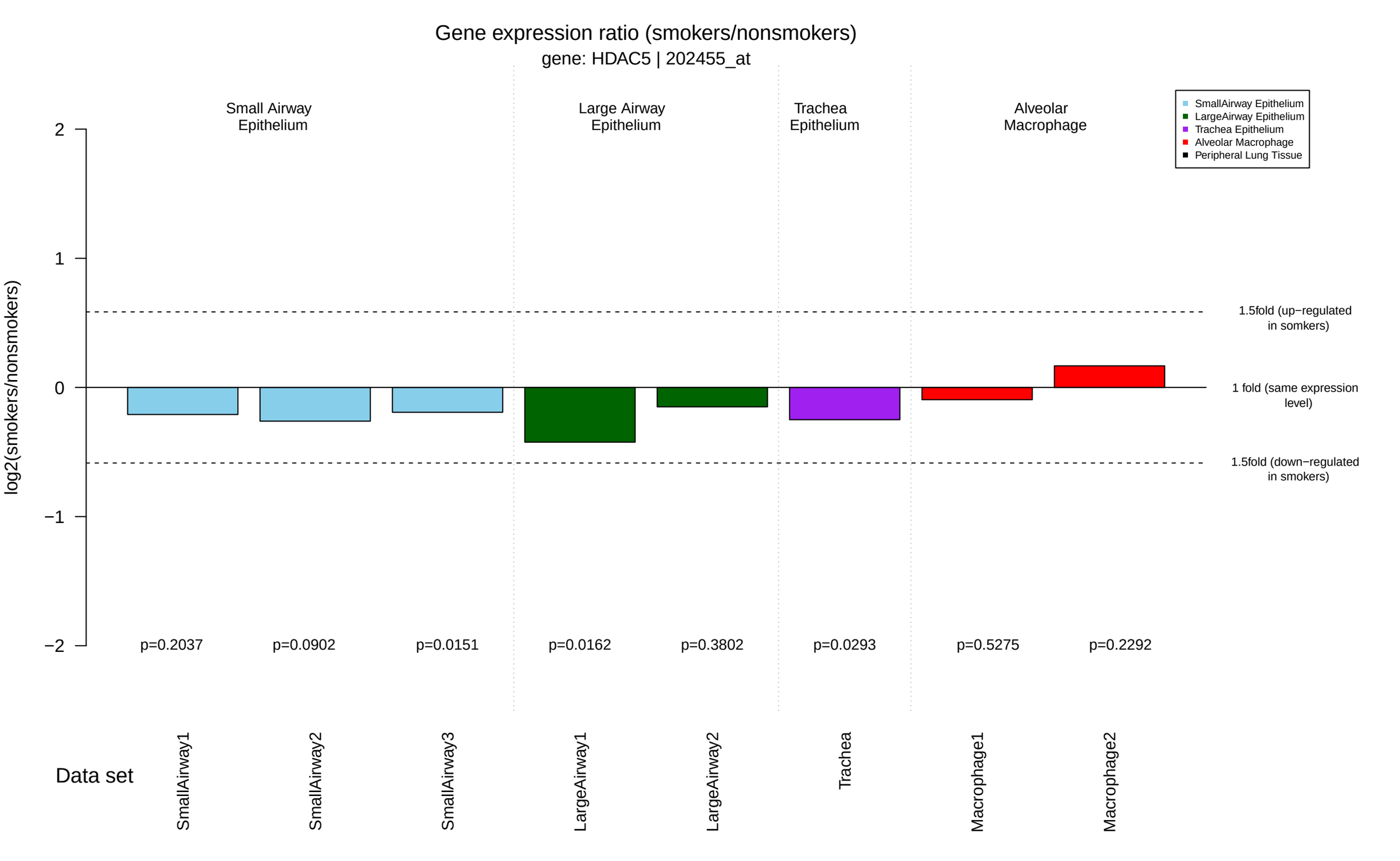
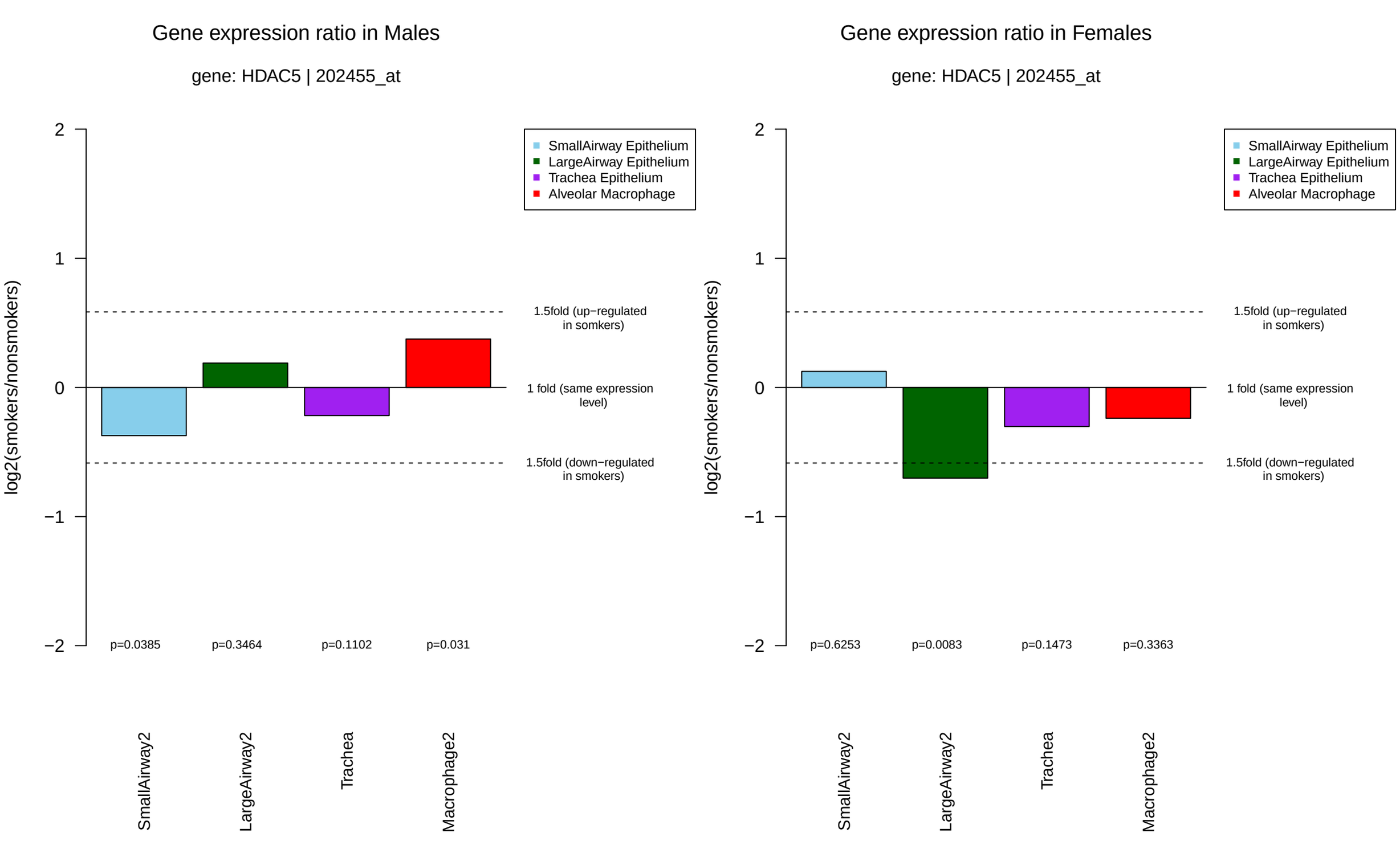
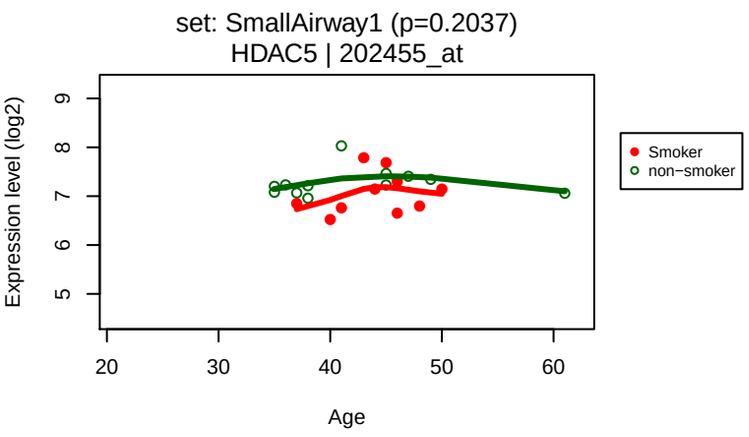
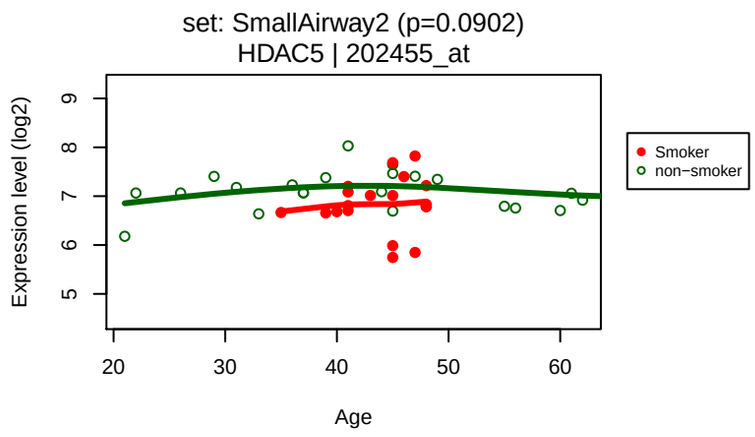
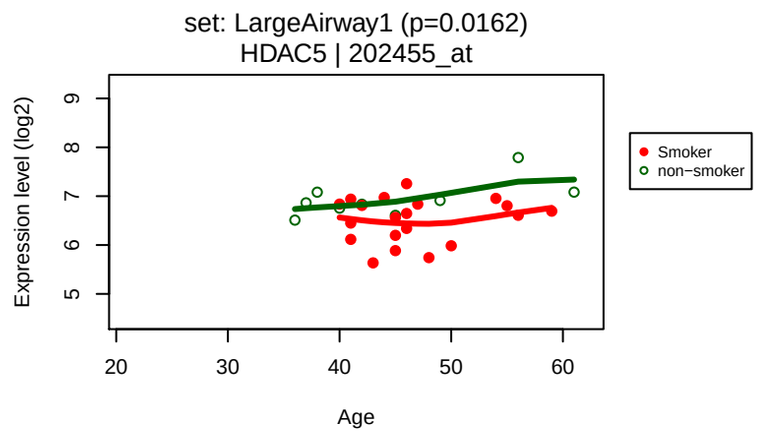

Gene expression of smokers and nonsmokers
gene: HDAC5 | 202455_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2037
p=0.0902
p=0.0151
p=0.0162
p=0.3802
p=0.0293
p=0.5275
p=0.2292
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

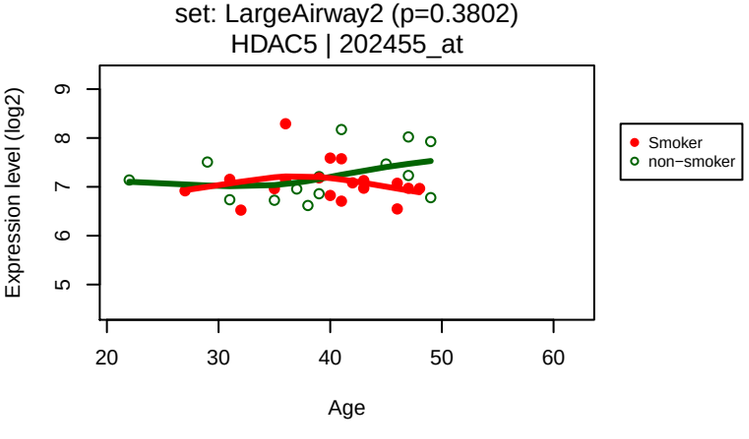

Gene expression ratio (smokers/nonsmokers)
gene: HDAC5 | 202455_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2037
p=0.0902
p=0.0151
p=0.0162
p=0.3802
p=0.0293
p=0.5275
p=0.2292
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

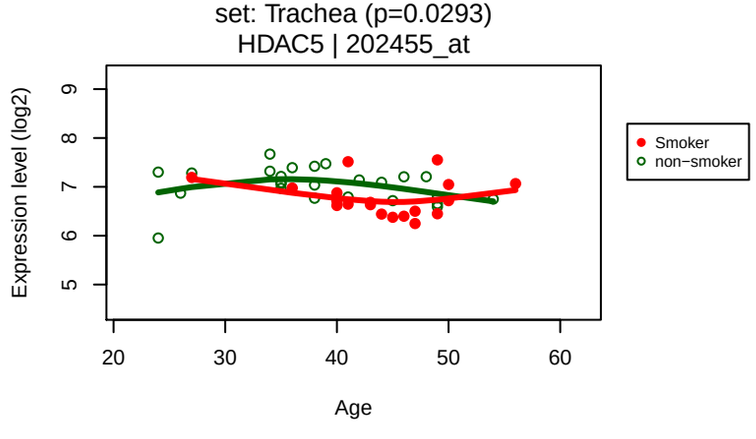

Gene expression ratio in Males
gene: HDAC5 | 202455_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0385
p=0.3464
p=0.1102
p=0.031
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HDAC5 | 202455_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6253
p=0.0083
p=0.1473
p=0.3363
SmallAirway2
LargeAirway2
Trachea
Macrophage2

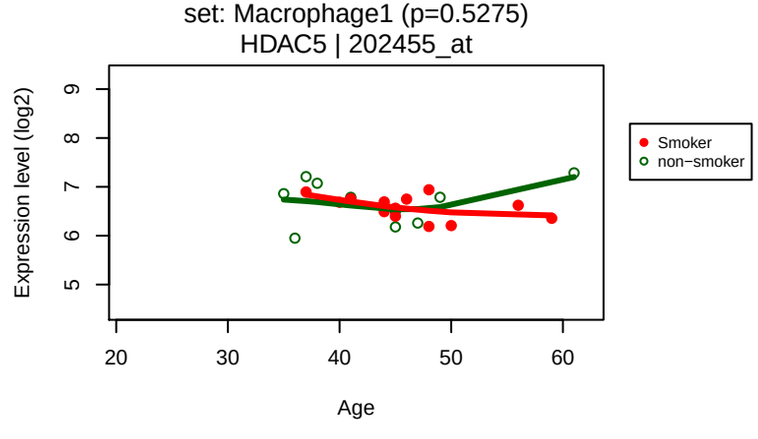

set: SmallAirway1 (p=0.2037)
HDAC5 | 202455_at
Expression level (log2)
Age
Smoker
non−smoker

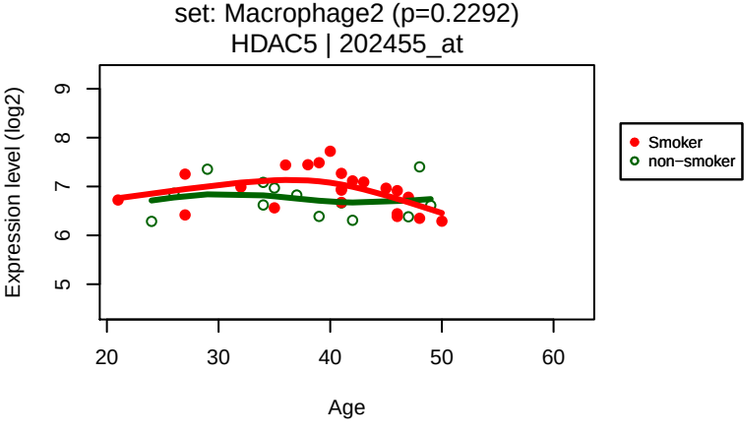

set: SmallAirway2 (p=0.0902)
HDAC5 | 202455_at
Expression level (log2)
Age
Smoker
non−smoker

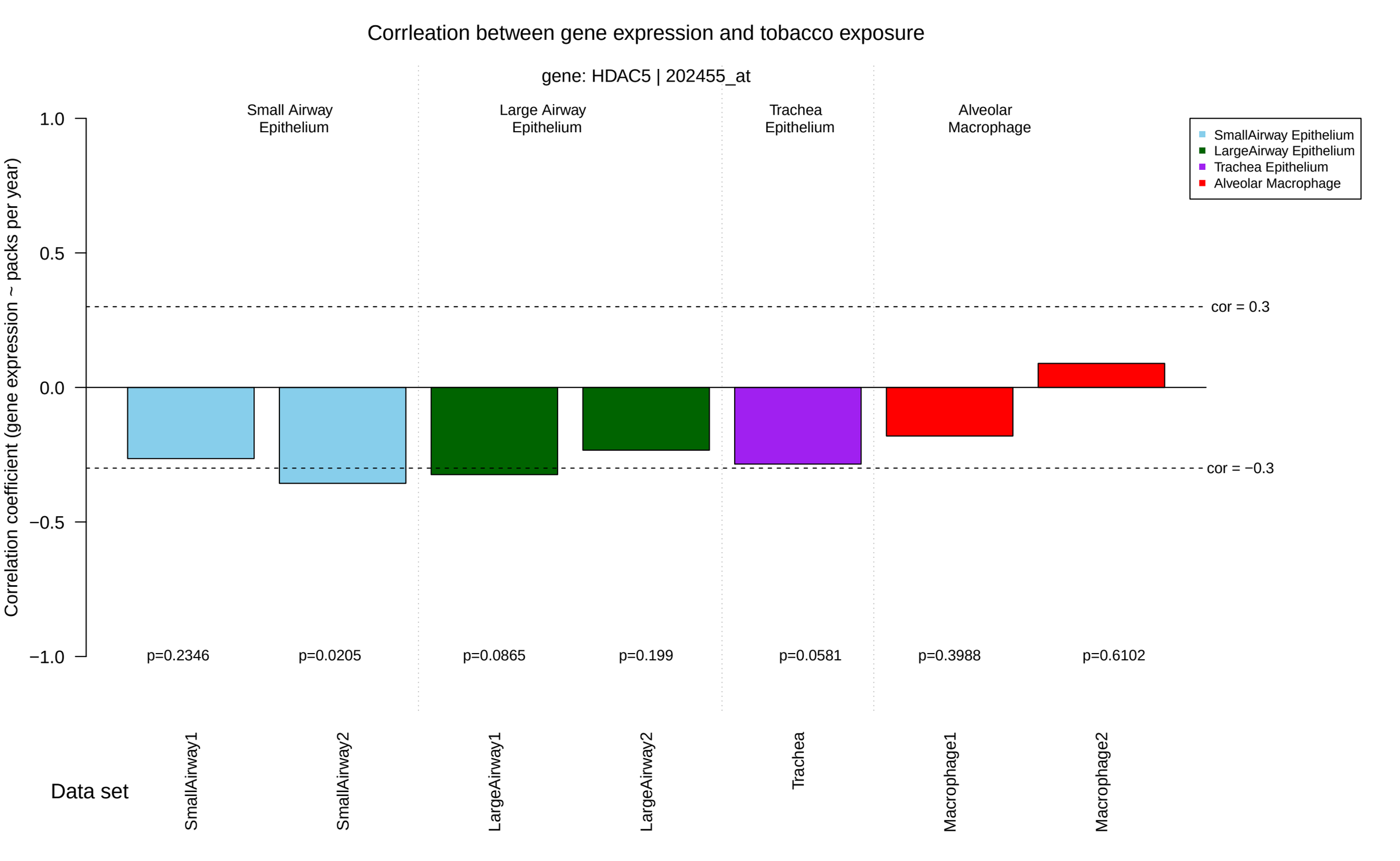

set: LargeAirway1 (p=0.0162)
HDAC5 | 202455_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3802)
HDAC5 | 202455_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0293)
HDAC5 | 202455_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5275)
HDAC5 | 202455_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2292)
HDAC5 | 202455_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HDAC5 | 202455_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2346
p=0.0205
p=0.0865
p=0.199
p=0.0581
p=0.3988
p=0.6102
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2